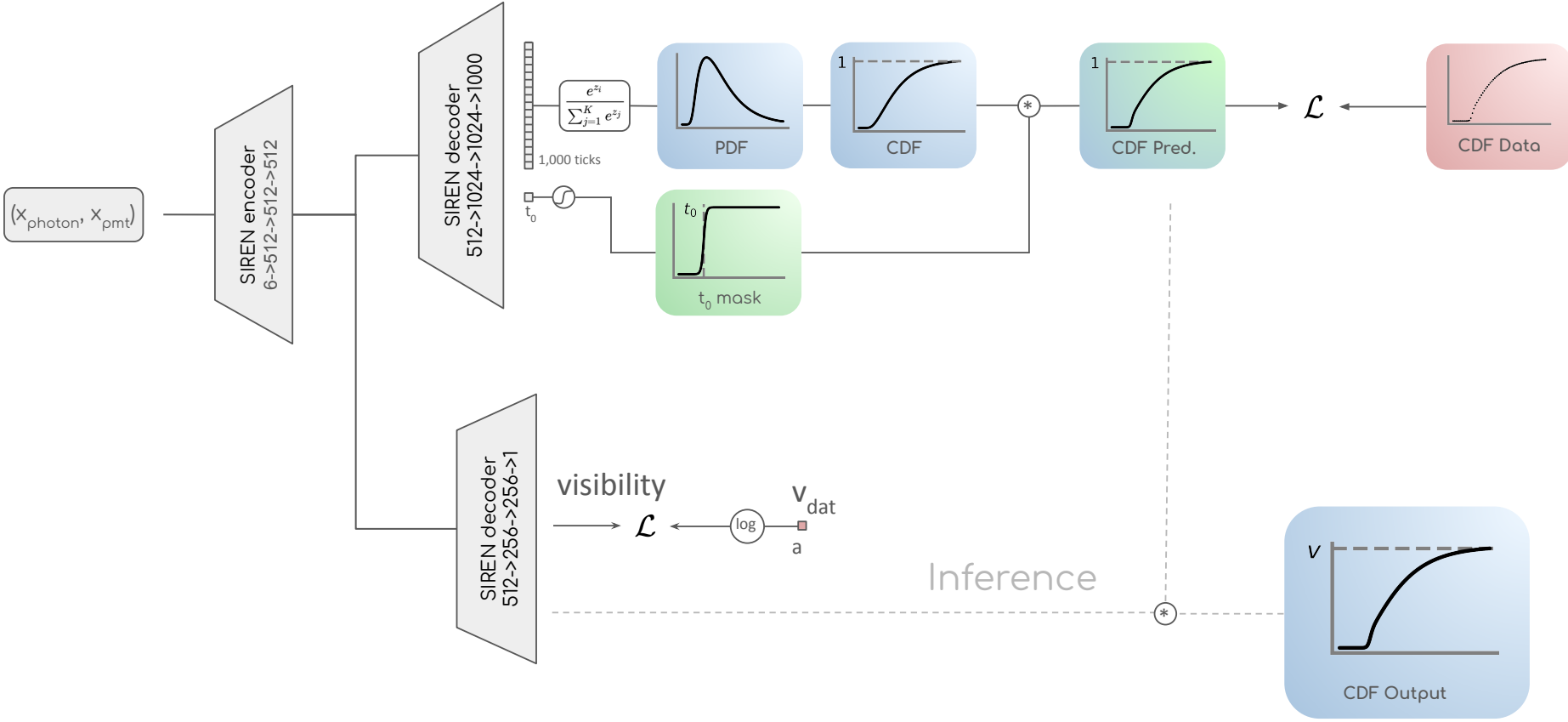


SIREN-TV

Sam and Junjie

cider-ml | Jan. 15, 2026

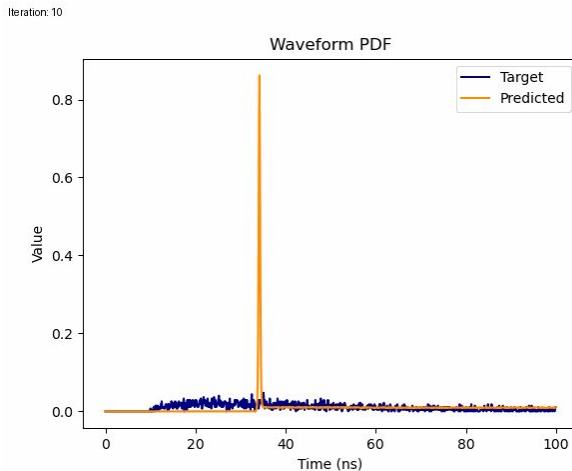
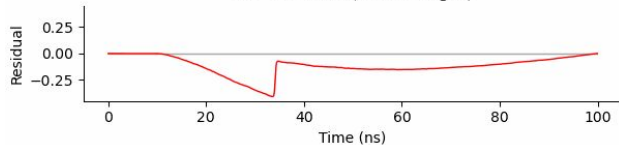
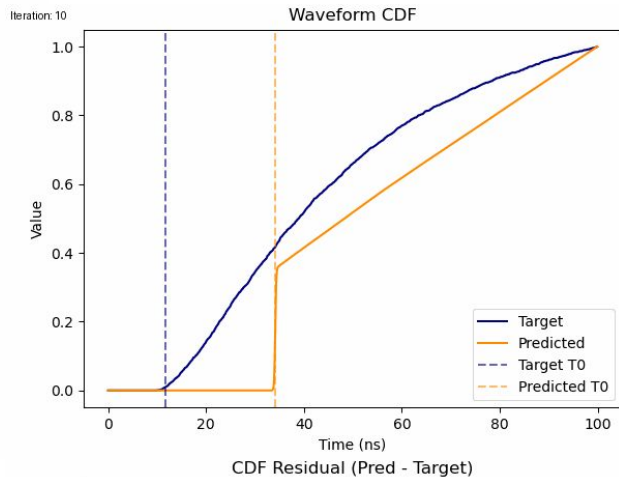
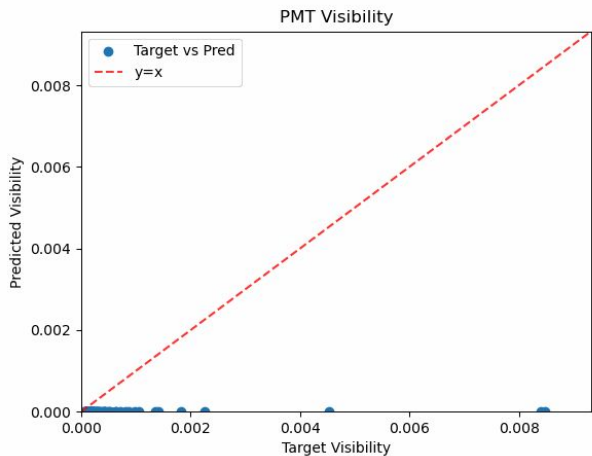
SirenTV – Branched



Training on ~174 voxels containing 10M photons

@ CM in Sep 2025

Iteration: 10

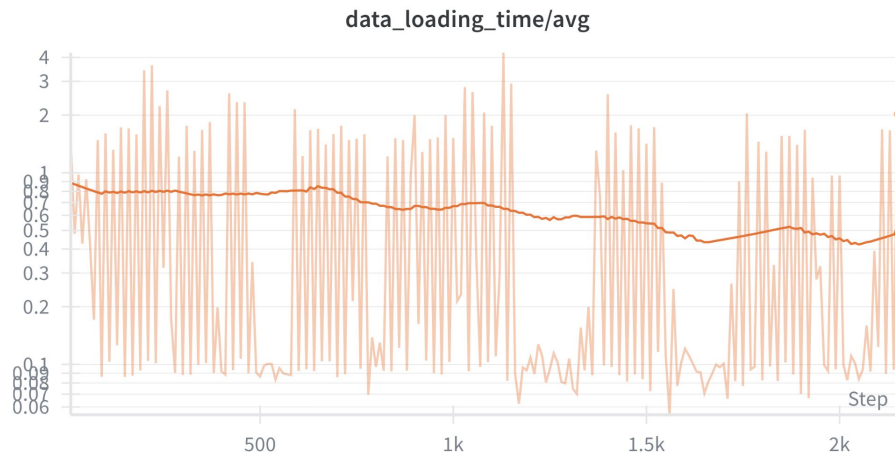


Expand to full dataset

- 1596672 voxels, 15M photons per voxel
- The old dataloader was a customized loader that didn't have the CPU multiprocessing implemented -> disk-memory transfer was ~4 sec per batch of 2048 voxels, while the network forward is ~0.35 s.

Rewrote the dataloader to fully inherit the features from Pytorch dataloader.

Time for 1 iteration, e.g. disk-CPU
and/or CPU-GPU transfer time



Used 4 workers in this test, can do more.

Expand to full dataset

- 1596672 voxels, 15M photons per voxel
- Enable distributed training, trains n times faster, where n = number of GPUs.



1 epoch = 195 iterations

Performance check

analysis/bias_vis/avg



analysis/bias_time/avg

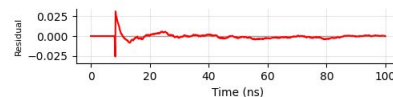
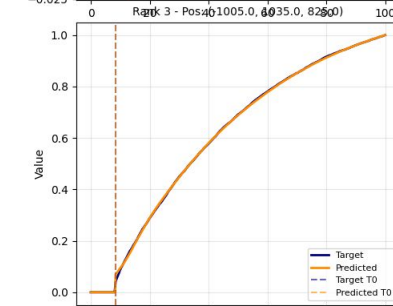
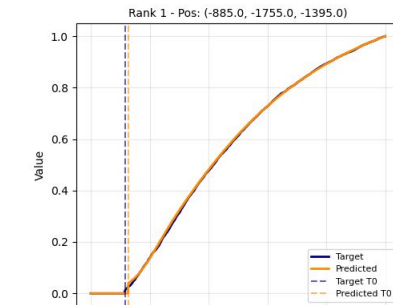
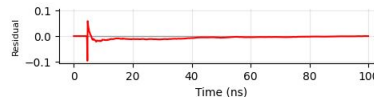
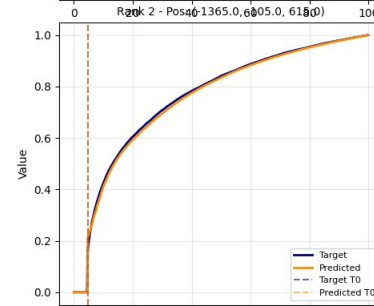
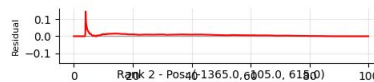
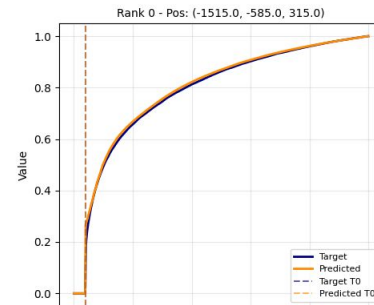
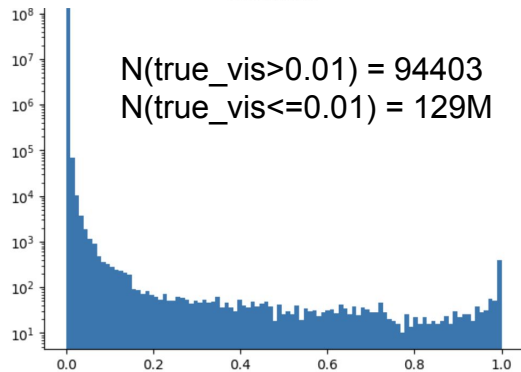
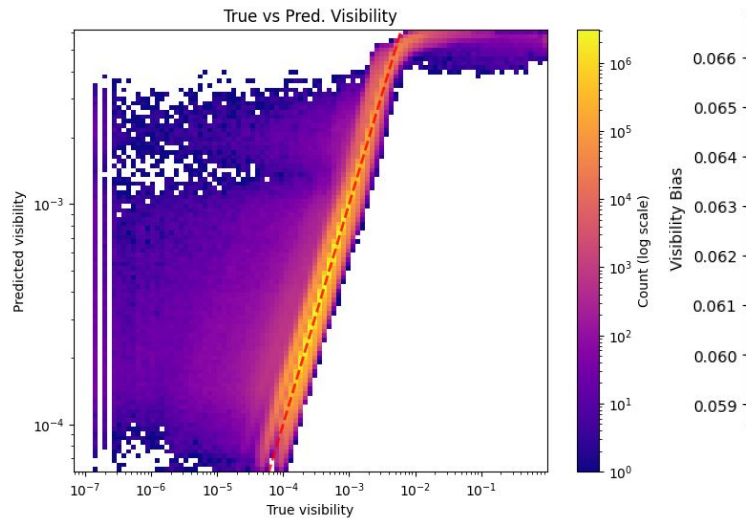


analysis/abs_bias_t0/avg



1 epoch = 195 iterations

Performance check @ epoch 10

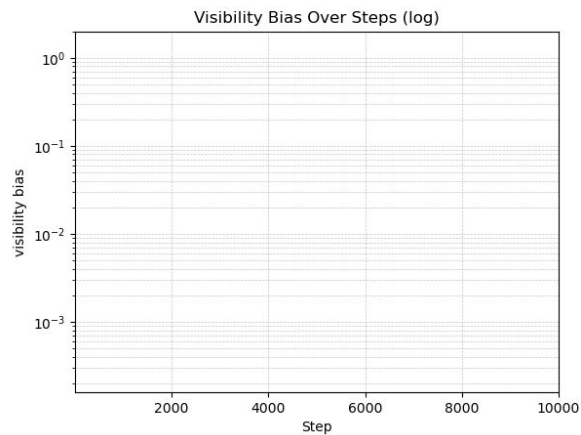


Backup

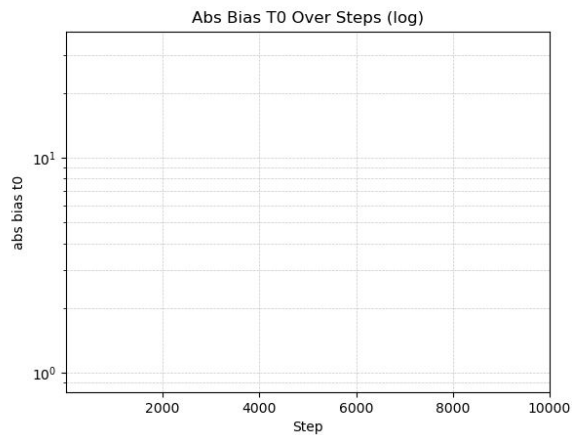
Bias

Fwd stats:

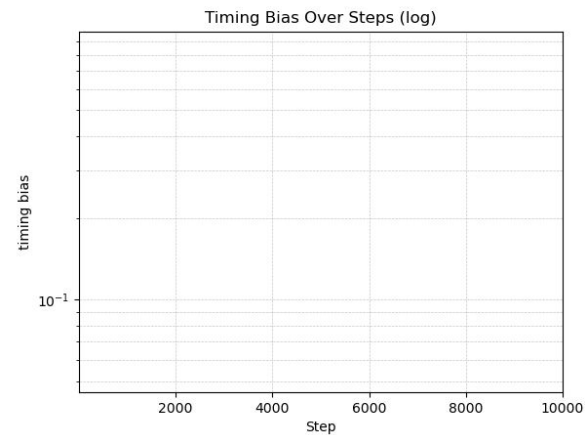
- 50 ms @ 1024 bsz, 5 GB VRAM
-



0.01%



0.85 ns



4.7%